

# Coordinate Geometry

Four formulas — midpoint, gradient, length, equation of a line — and one rule: perpendicular gradients multiply to  $-1$ .



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## THE TOPIC ON ONE MAP

### Coordinate Geometry

#### 01 — Midpoint and length

midpoint of  $AB$

find  $B$  from the midpoint

#### 02 — Gradient and the equation of a line

gradient of  $2y + 7x = 10$

through  $A$  and  $B$

#### 03 — Parallel and perpendicular

$m_1 m_2 = -1$

perpendicular bisector

tangent to a circle

#### 04 — Shapes on a grid

rectangle, kite, rhombus

area of triangle  $AOB$

#### 05 — Line meets curve

substitute, solve

midpoint of  $PQ$

exact length  $AB$

**THE TOOLKIT** Four formulas for two points  $(x_1, y_1)$  and  $(x_2, y_2)$ :

#### ZONE MID

##### Midpoint

$$\left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

#### ZONE m

##### Gradient

$$\frac{y_2 - y_1}{x_2 - x_1}$$

#### ZONE LEN

##### Length

$$\sqrt{(\Delta x)^2 + (\Delta y)^2}$$

#### ZONE LINE

##### Equation

$$y - y_1 = m(x - x_1)$$

# 01 Midpoint and Length

Midpoint: average the  $x$ 's, average the  $y$ 's. Length: Pythagoras on the differences. To find an end from the midpoint, go the same step again.

## WHEN YOU SEE → YOU DO

**"the midpoint of  $AB$ "**

→ average each coordinate

**"the midpoint of  $AB$  is  $(a, -19)$ , find  $b$ "**

$$\rightarrow \frac{11 + b}{2} = -19$$

**" $M$  is the midpoint, find  $B$ "**

→  $B = 2M - A$  (step from  $A$  to  $M$ , then again)

**"the distance between the centres"**

$$\rightarrow \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

**" $T$  divides  $OP$  in the ratio  $1 : 2$ "**

$$\rightarrow T = O + \frac{1}{3}(P - O)$$

## WORKED MODEL • FIND AN END

$P(5, 7)$  and the midpoint of  $PQ$  is  $M(1, 2.5)$ . Find  $Q$ .

**1** step  $P \rightarrow M$ :  $(-4, -4.5)$

how far

**2**  $Q = M + (-4, -4.5)$

same step again

**ANSWER**  $Q(-3, -2)$

IDEA 01

## Practise: Midpoint and Length

My questions first, then selected practice, then real past papers.

**Q1** DR ESLAM Midpoint

CH 3 • TOPIC 2 2 marks

$A(-3, 5)$ ,  $B(7, -1)$ . Find the midpoint of  $AB$  and the length of  $AB$ .

## SELECTED PRACTICE

Q2

PRACTICE

Practice • Easy

CH 3 • TOPIC 2

2 marks

- 1 The point  $A$  has coordinates  $(2, 3)$ .  
The point  $B$  has coordinates  $(6, 8)$ .

$M$  is the midpoint of the line  $AB$ .  
Find the coordinates of  $M$ .

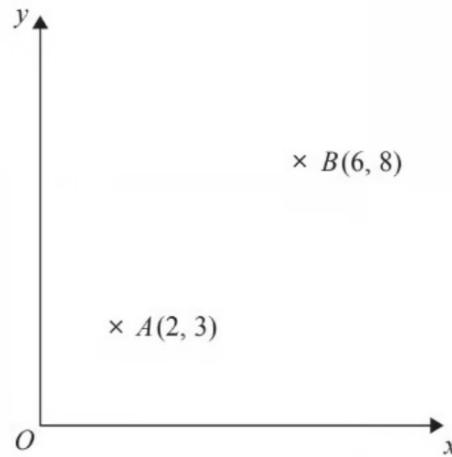


Diagram **NOT**  
accurately drawn

(2 marks)

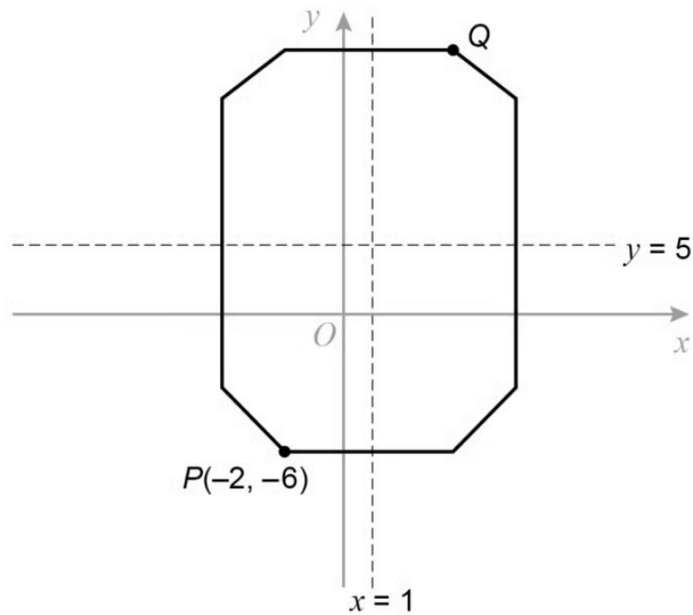
**Q3****PRACTICE**

Practice · Easy

CH 3 · TOPIC 2

**2 marks**

11 The diagram shows an octagon



Not drawn  
accurately

$x = 1$  and  $y = 5$  are lines of symmetry.

Work out the coordinates of point  $Q$ .

**(2 marks)**

**Q4****PRACTICE**

Practice • Medium

CH 3 • TOPIC 2

**3 marks**

6 The points  $A$ ,  $B$  and  $C$  lie in order on a straight line.

The coordinates of  $A$  are  $(2, 5)$

The coordinates of  $B$  are  $(4, p)$

The coordinates of  $C$  are  $(q, 17)$

Given that  $AC = 4AB$ , find the values of  $p$  and  $q$ .

**(3 marks)**

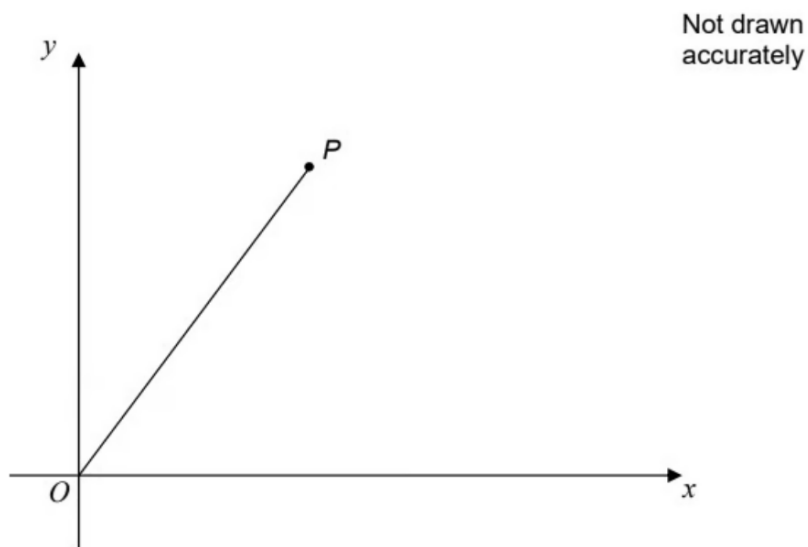
**Q5****PRACTICE**

Practice · Hard

CH 3 · TOPIC 2

4 marks

- 7 The diagram shows a line joining  $O$  to  $P$ .



The gradient of the line is 2

The length of the line is  $\sqrt{2645}$

Work out the coordinates of  $P$ .

(4 marks)

## PAST PAPERS

**Q6****PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q6

**CH 3 • TOPIC 2 2 marks**

- 6 Point  $A$  has coordinates  $(-3, 11)$   
Point  $B$  has coordinates  $(47, b)$   
The midpoint of  $AB$  has coordinates  $(a, -19)$

Find the value of  $a$  and the value of  $b$ .

 $a = \dots\dots\dots$  $b = \dots\dots\dots$ 

(Total for Question 6 is 2 marks)

**Q7****PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q10

**CH 3 • TOPIC 2 4 marks**

- 10** Two circles,  $C_1$  and  $C_2$ , are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle  $C_1$  is at the point with coordinates  $(-1, 3)$  and the radius of  $C_1$  is 13 cm.

The centre of circle  $C_2$  is at the point with coordinates  $(7, 18)$  and the radius of  $C_2$  is 6 cm.

- (a) Work out the distance between the centre of  $C_1$  and the centre of  $C_2$

..... cm

- (b) Explain why circle  $C_1$  intersects circle  $C_2$

.....  
 .....

**(Total for Question 10 is 4 marks)**

**Q8****PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q1

**CH 3 · TOPIC 2****2 marks**

**You must write down all the stages in your working.**

- 1** The point  $P$  has coordinates  $(3, 4)$   
The point  $Q$  has coordinates  $(9, 16)$

$M$  is the midpoint of the line  $PQ$

Find the coordinates of  $M$

(..... , .....)

**(Total for Question 1 is 2 marks)**

## 02 Gradient and the Equation of a Line

Gradient = rise over run. For an equation, make  $y$  the subject to read  $m$  and  $c$ . For a line through two points: gradient first, then  $y - y_1 = m(x - x_1)$ .

### WHEN YOU SEE → YOU DO

“the gradient of  $2y + 7x = 10$ ”

$$\rightarrow y = -\frac{7}{2}x + 5: m = -\frac{7}{2}$$

“crosses the  $y$ -axis”

$$\rightarrow \text{put } x = 0$$

“the equation of the line through A and B”

$$\rightarrow m \text{ first, then } y - y_1 = m(x - x_1)$$

“in the form  $ax + by + c = 0$  with integers”

$$\rightarrow \text{clear fractions, all on one side}$$

“does the point lie on the line?”

$$\rightarrow \text{substitute } x, \text{ compare } y$$

Gradient

Substitute a point

Rearrange to the form asked

### WORKED MODEL • MAY 2023 P1H

Find the equation of the line through  $A(-5, 12)$  and  $B(19, -48)$ .

$$1 \quad m = \frac{-48 - 12}{19 - (-5)} = -\frac{60}{24} = -\frac{5}{2}$$

gradient

$$2 \quad y - 12 = -\frac{5}{2}(x + 5)$$

point A

ANSWER  $y = -\frac{5}{2}x - \frac{1}{2}$

IDEA 02

## Practise: Gradient and the Equation of a Line

My questions first, then selected practice, then real past papers.

**Q9** **DR ESLAM** Line through two points

CH 3 • TOPIC 2 3 marks

Find the equation of the line through (1, 3) and (4, 12).

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### SELECTED PRACTICE

**Q10** **PRACTICE** Practice • Easy

CH 3 • TOPIC 2 2 marks

5 Find the gradient of the straight line with equation  $5x + 2y = 7$

(2 marks)

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**Q11****PRACTICE**

Practice · Easy

CH 3 · TOPIC 2

2 marks

10 Work out the gradient of the straight line through  $(-2, 3)$  and  $(1, 9)$

(2 marks)

**Q12****PRACTICE**

Practice · Hard

CH 3 · TOPIC 2

5 marks

1 (a)  $A$  and  $B$  are two points.

Point  $A$  has coordinates  $(-2, 4)$ .

Point  $B$  has coordinates  $(8, 9)$ .

$C$  is the midpoint of the line segment  $AB$ .

Find the coordinates of  $C$ .

(2 marks)

(b)  $D$  is the point with coordinates  $(100, 56)$ .

Does point  $D$  lie on the straight line that passes through  $A$  and  $B$ ?

You must show how you work out your answer.

(3 marks)

**Q13****PRACTICE**

Practice · Hard

CH 3 • TOPIC 2 3 marks

9 A straight line goes through the points  $(p, q)$  and  $(r, s)$ , where

- $p + 2 = r$
- $q + 4 = s.$

Find the gradient of the line.

(3 marks)

## PAST PAPERS

**Q14** PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1H • Q1

CH 3 • TOPIC 2 4 marks

**You must write down all the stages in your working.**

- 1 The point  $A$  has coordinates  $(5, -4)$   
The point  $B$  has coordinates  $(13, 1)$

(a) Work out the coordinates of the midpoint of  $AB$ .

(....., .....)

Line  $L$  has equation  $y = 2 - 3x$

(b) Write down the gradient of line  $L$ .

.....

Line  $L$  has equation  $y = 2 - 3x$

- (c) Does the point with coordinates  $(100, -302)$  lie on line  $L$ ?  
You must give a reason for your answer.

.....  
.....  
.....

**(Total for Question 1 is 4 marks)**

**Q15****PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q13

**CH 3 • TOPIC 2 3 marks**

- 13 Point  $A$  has coordinates  $(5, 8)$   
Point  $B$  has coordinates  $(9, -4)$

(a) Work out the gradient of  $AB$ .

.....

The straight line  $L$  has equation  $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to  $L$ .

.....

**(Total for Question 13 is 3 marks)**

**Q16****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q10

**CH 3 • TOPIC 2 3 marks**

- 10 The straight line  $L$  has equation  $2y + 7x = 10$

(a) Find the gradient of  $L$

.....

(b) Find the coordinates of the point where  $L$  crosses the  $y$ -axis.

(....., .....)

**(Total for Question 10 is 3 marks)**

**Q17** **PAST PAPER** Edexcel 4MA1 · May 2023 · Paper 1H · Q13**CH 3 · TOPIC 2** **3 marks**

13  $A$  is the point with coordinates  $(-5, 12)$

$B$  is the point with coordinates  $(19, -48)$

Find an equation of the straight line that passes through the points  $A$  and  $B$

.....

**(Total for Question 13 is 3 marks)**

## 03 Parallel and Perpendicular

Parallel: same gradient. Perpendicular: flip and change sign ( $m_1 m_2 = -1$ ). The perpendicular bisector goes through the midpoint.

### WHEN YOU SEE → YOU DO

“parallel to  $y = 7 - 4x$ ”

→ same  $m = -4$

“perpendicular to  $L$ ”

→  $-\frac{1}{m}: \frac{2}{5} \rightarrow -\frac{5}{2}$

“perpendicular bisector of  $PQ$ ”

→ midpoint + perpendicular gradient

“tangent to the circle at  $P$ ”

→ perpendicular to the radius  $OP$

“show that angle  $PQR$  is not a right angle”

→  $m_1 m_2 \neq -1$

### ZONE PARALLEL

$$m_1 = m_2$$



### ZONE PERPENDICULAR

$$m_1 \times m_2 = -1$$



### WORKED MODEL • JUNE 2025 P2HR

$P(-5, 2)$ ,  $Q(7, 6)$ . Find the perpendicular bisector of  $PQ$  as  $y = mx + c$ .

1 midpoint  $(1, 4)$ ;  $m_{PQ} = \frac{4}{12} = \frac{1}{3}$

midpoint, gradient

2  $m = -3$ :  $y - 4 = -3(x - 1)$

perpendicular

ANSWER  $y = -3x + 7$

## IDEA 03

**Practise: Parallel and Perpendicular**

My questions first, then selected practice, then real past papers.

**Q18** DR ESLAM Perpendicular

CH 3 • TOPIC 2 3 marks

Find the line perpendicular to  $y = \frac{1}{2}x + 4$  through  $(2, 1)$ .

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**SELECTED PRACTICE****Q19** PRACTICE Practice • Medium

CH 3 • TOPIC 2 3 marks

- 7  $P$  is the point  $(2, 14)$   
 $Q$  is the point  $(6, 8)$   
 $R$  is the point  $(2, 5)$

Use gradients to show that angle  $PQR$  is not a right angle.

**(3 marks)**

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**Q20****PRACTICE**

Practice • Hard

**CH 3 • TOPIC 2****4 marks**

**3** A triangle has vertices  $P$ ,  $Q$  and  $R$ .

The coordinates of  $P$  are  $(-3, -6)$

The coordinates of  $Q$  are  $(1, 4)$

The coordinates of  $R$  are  $(5, -2)$

$M$  is the midpoint of  $PQ$ .

$N$  is the midpoint of  $QR$ .

Prove that  $MN$  is parallel to  $PR$ .

You must show each stage of your working.

**(4 marks)**

## PAST PAPERS

**Q21****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q19

**CH 3 • TOPIC 2 5 marks**

- 19 (a) Write down an equation of a line that is parallel to the line with equation  $y = 7 - 4x$

The line **L** passes through the points with coordinates  $(-3, 1)$  and  $(2, -2)$

- (b) Find an equation of the line that is perpendicular to **L** and passes through the point with coordinates  $(-6, 4)$

Give your answer in the form  $ax + by + c = 0$  where  $a$ ,  $b$  and  $c$  are integers.

**(Total for Question 19 is 5 marks)**

**Q22****PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q17

**CH 3 • TOPIC 2 4 marks**

17 The straight line **L** passes through the points  $(4, -1)$  and  $(6, 4)$

The straight line **M** is perpendicular to **L** and intersects the  $y$ -axis at the point  $(0, 8)$

Find the coordinates of the point where **M** intersects the  $x$ -axis.

(....., .....)

(Total for Question 17 is 4 marks)

**Q23****PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q20

**CH 3 • TOPIC 2 4 marks**

20 The centre  $O$  of a circle has coordinates  $(4, 7)$

The point  $A$ , on the circle, has coordinates  $(6, 11)$  and  $AOP$  is a diameter of the circle.

Find an equation of the tangent to the circle at the point  $P$

.....

(Total for Question 20 is 4 marks)

**Q24****PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q25

**CH 3 • TOPIC 2 6 marks**

- 25** The straight line with equation  $y - 2x = 7$  is the perpendicular bisector of the line  $AB$  where  $A$  is the point with coordinates  $(j, 7)$  and  $B$  is the point with coordinates  $(6, k)$

Find the coordinates of the midpoint of the line  $AB$

Show clear algebraic working.

(..... , .....)

**(Total for Question 25 is 6 marks)**

**Q25****PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q22

**CH 3 · TOPIC 2 6 marks**

- 22 [In this question 1 cm = 1 unit on the  $x$ -axis and  
1 cm = 1 unit on the  $y$ -axis]

$P$  is a point on a circle with centre  $(0, 0)$

The coordinates of  $P$  are  $(8, -10)$

The line  $L$  is the tangent to the circle at the point  $P$

$L$  crosses the  $x$ -axis at the point  $Q$  and crosses the  $y$ -axis at the point  $R$

Work out the length of  $RQ$

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

**Q26****PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q24

**CH 3 · TOPIC 2 6 marks**

- 24  $PQ$  is a straight line drawn on a square grid, with a scale of 1 cm for 1 unit on each axis.

$P$  has coordinates  $(-5, a)$  and  $Q$  has coordinates  $(7, 3a)$  where  $a > 0$

The length of  $PQ$  is  $4\sqrt{10}$  cm

Find an equation of the perpendicular bisector of  $PQ$

Give your answer in the form  $y = mx + c$  where  $m$  and  $c$  are integers.

(Total for Question 24 is 6 marks)

**Q27****PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q20

**CH 3 • TOPIC 2 3 marks**

**20** The straight line **L** has equation  $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form  $y = mx + c$

.....  
(Total for Question 20 is 3 marks)

## 04 Shapes on a Grid

Sketch it. Use the shape's facts: opposite sides parallel (same gradient), a right angle ( $m_1 m_2 = -1$ ), diagonals bisect each other (same midpoint), equal lengths.

### WHEN YOU SEE → YOU DO

**rectangle  $ABCD$ , one coordinate unknown**

→  $AB \perp BC$ : gradients multiply to  $-1$

**rhombus / kite**

→ diagonals are perpendicular; a kite is symmetric in one diagonal

**"area of triangle  $AOB$ "**

→ find where the lines cross the axes

**"isosceles,  $HJ = HK$ "**

→  $H$  is on the perpendicular bisector of  $JK$

### WORKED MODEL • JANUARY 2021 P2H

Rectangle:  $A(-4, -2)$ ,  $B(1, 10)$ ,  $C(19, a)$ . Find  $a$ .

1  $m_{AB} = \frac{12}{5}$ , so  $m_{BC} = -\frac{5}{12}$

right angle at  $B$

2  $\frac{a - 10}{18} = -\frac{5}{12}$

gradient of  $BC$

**ANSWER**  $a = 2.5$

IDEA 04

## Practise: Shapes on a Grid

My questions first, then selected practice, then real past papers.

**Q28** **DR ESLAM** Right angle

CH 3 • TOPIC 2 3 marks

$A(0, 0)$ ,  $B(4, 2)$ ,  $C(k, 8)$ . Angle  $ABC = 90^\circ$ . Find  $k$ .

## SELECTED PRACTICE

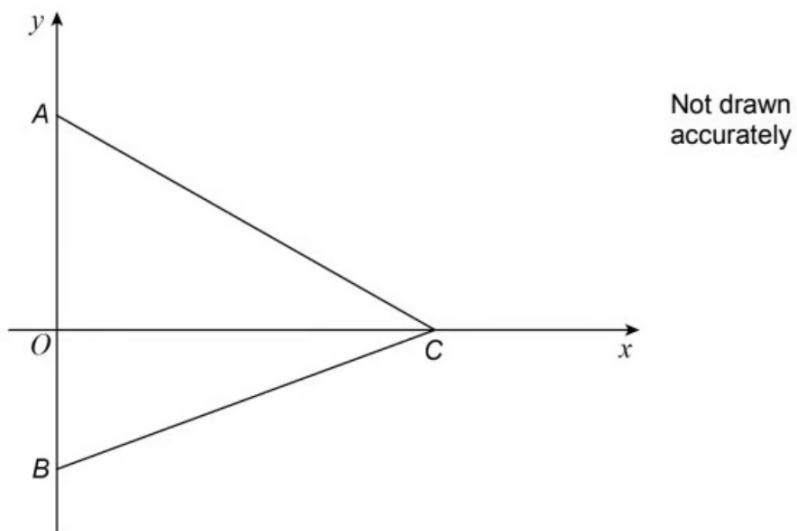
**Q29**

PRACTICE

Practice • Medium

CH 3 • TOPIC 2 2 marks

10  $A$ ,  $B$  and  $C$  are points on the axes as shown.



The area of triangle  $ABC$  is 28 square units.

Work out possible coordinates for  $A$ ,  $B$  and  $C$ .

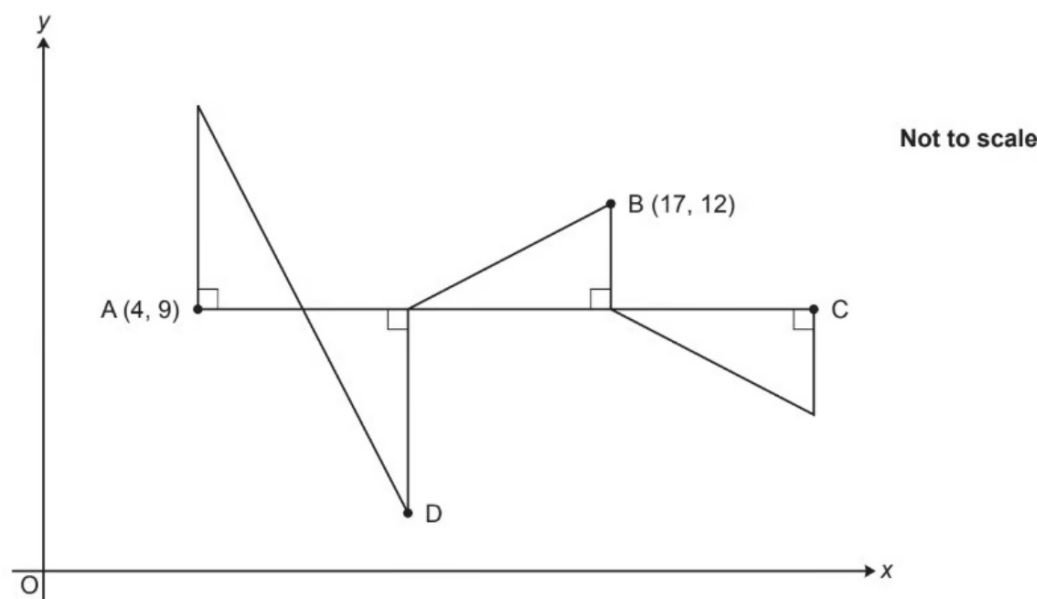
 $A$  ( ..... , ..... ) $B$  ( ..... , ..... ) $C$  ( ..... , ..... )  
(2 marks)

**Q30****PRACTICE**

Practice • Hard

**CH 3 • TOPIC 2 5 marks**

8 A pattern is made from four congruent right-angled triangles.



The line AC is parallel to the x - axis.

The point A has coordinates (4, 9) and the point B has coordinates (17, 12).

Work out the coordinates of point C and point D.

$C$  ( ..... , ..... )

$D$  ( ..... , ..... )

(5 marks)

## PAST PAPERS

**Q31****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q22

**CH 3 • TOPIC 2 6 marks**

22 Triangle  $HJK$  is isosceles with  $HJ = HK$  and  $JK = \sqrt{80}$

$H$  is the point with coordinates  $(-4, 1)$

$J$  is the point with coordinates  $(j, 15)$  where  $j < 0$

$K$  is the point with coordinates  $(6, k)$

$M$  is the midpoint of  $JK$ .

The gradient of  $HM$  is 2

Find the value of  $j$  and the value of  $k$ .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

**(Total for Question 22 is 6 marks)**

**Q32****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q24

**CH 3 • TOPIC 2 5 marks**

24  $L_1$  and  $L_2$  are two straight lines.

The origin of the coordinate axes is  $O$ .

$L_1$  has equation  $5x + 10y = 8$

$L_2$  is perpendicular to  $L_1$  and passes through the point with coordinates  $(8, 6)$

$L_2$  crosses the  $x$ -axis at the point  $A$ .

$L_2$  intersects the straight line with equation  $x = -3$  at the point  $B$ .

Find the area of triangle  $AOB$ .

Show your working clearly.

.....  
(Total for Question 24 is 5 marks)

**Q33****PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q22

**CH 3 • TOPIC 2 4 marks**

**22**  $ABCD$  is a rhombus.

The diagonals,  $AC$  and  $BD$ , intersect at the point  $M$ .

The coordinates of  $M$  are  $(6, -11)$

The points  $A$  and  $C$  both lie on the line with equation  $2y + 7x = 20$

Find the exact coordinates of the point where the line through  $B$  and  $D$  intersects the  $y$ -axis.

(....., .....)

**(Total for Question 22 is 4 marks)**

**Q34** **PAST PAPER** Edexcel 4MA1 • January 2021 • Paper 1HR • Q22**CH 3 • TOPIC 2** **5 marks**

**22**  $ABC$  is an isosceles triangle with  $AB = AC$ .

$B$  is the point with coordinates  $(-1, 5)$

$C$  is the point with coordinates  $(2, 10)$

$M$  is the midpoint of  $BC$ .

Find an equation of the line through the points  $A$  and  $M$ .

Give your answer in the form  $py + qx = r$  where  $p$ ,  $q$  and  $r$  are integers.

(Total for Question 22 is 5 marks)

**Q35****PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2H · Q18

**CH 3 · TOPIC 2 7 marks**

18 A rectangle  $ABCD$  is to be drawn on a centimetre grid such that

$A$  has coordinates  $(-4, -2)$

$B$  has coordinates  $(1, 10)$

$C$  has coordinates  $(19, a)$

$D$  has coordinates  $(b, c)$

(a) Work out the value of  $a$ , the value of  $b$  and the value of  $c$ .

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle  $ABCD$ .

$\dots\dots\dots$  cm

**(Total for Question 18 is 7 marks)**

**Q36****PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q20

**CH 3 • TOPIC 2 7 marks**

- 20** The straight line **L** passes through point  $A(-6, 2)$  and point  $B(5, 3)$   
The straight line **M** is perpendicular to **L** and passes through the midpoint of  $A$  and  $B$ .  
The line **M** intersects the line  $x = -1$  at point  $C$ .

Calculate the area of triangle  $ABC$ .

.....  
(Total for Question 20 is 7 marks)

**Q37****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q22

**CH 3 • TOPIC 2 5 marks**

**22**  $ABC$  is a triangle in which angle  $ABC = 90^\circ$

$p$  and  $q$  are integers such that

the coordinates of  $A$  are  $(p, 10)$

the coordinates of  $B$  are  $(-1, -5)$

the coordinates of  $C$  are  $(8, q)$

Given that the gradient of  $AC$  is  $-\frac{6}{7}$

work out the value of  $p$  and the value of  $q$

$p = \dots\dots\dots$

$q = \dots\dots\dots$

**(Total for Question 22 is 5 marks)**

**Q38****PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2H · Q22

**CH 3 · TOPIC 2 7 marks**

**22**  $ABCD$  is a kite, with diagonals  $AC$  and  $BD$ , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

$A$  is the point with coordinates  $(-3, 4)$

The diagonals of the kite intersect at the point  $M$  with coordinates  $(0, 2)$

Given that  $AB = AD = 6.5$  cm and the  $x$  coordinate of  $B$  is positive,

find the coordinates of the points  $B$  and  $D$ .

(..... , .....)

(..... , .....)

(Total for Question 22 is 7 marks)

**Q39****PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2H · Q24

**CH 3 · TOPIC 2 6 marks**

24  $ABCD$  is a kite with  $AB = AD$  and  $CB = CD$

$A$  is the point with coordinates  $(-2, 10)$

$B$  is the point with coordinates  $\left(-\frac{27}{5}, 4\right)$

$C$  is the point with coordinates  $(4, -5)$

Work out the coordinates of  $D$

(..... , .....)

(Total for Question 24 is 6 marks)

**Q40****PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q21

**CH 3 • TOPIC 2 4 marks**

21  $ABCD$  is a square.

The point  $A$  has coordinates  $(-5, 2)$

The point  $B$  has coordinates  $(3, 5)$

Find an equation of the line that passes through  $B$  and  $C$

Give your answer in the form  $ax + by + c = 0$  where  $a$ ,  $b$  and  $c$  are integers.

.....

**(Total for Question 21 is 4 marks)**

**Q41** PAST PAPER Edexcel 4MA1 • June 2025 • Paper 1H • Q25

CH 3 • TOPIC 2 5 marks

25  $PQRS$  is a square.

$PR$  is a diagonal of the square.

$P$  is the point with coordinates  $(4, 7)$

$R$  is the point with coordinates  $(8, -5)$

Find an equation of the straight line that passes through the points  $Q$  and  $S$

Give your answer in the form  $ay = bx + c$  where  $a$ ,  $b$  and  $c$  are integers.

(Total for Question 25 is 5 marks)

## 05 Line Meets Curve

Solve them simultaneously (Topic 2.12): substitute the line into the curve, solve the quadratic, pair the coordinates. Then midpoint or length.

### WHEN YOU SEE → YOU DO

“**L and C intersect at P and Q**”

→ substitute, solve, two points

“**the midpoint of PQ**”

→ average the two points

“**the exact length of AB**”

→ Pythagoras, keep the surd

“**one point A is given, find B**”

→ use A to find the unknown constants first

### WORKED MODEL • NOVEMBER 2025 P1H

$x^2 + y^2 = d - 11x$  and  $y = 3x + e$  meet at  $A(0.2, 2.6)$  and  $B$ . Find  $B$ .

1  $2.6 = 0.6 + e \Rightarrow e = 2; 0.04 + 6.76 = d - 2.2 \Rightarrow d = 9$

use A

2  $x^2 + (3x + 2)^2 = 9 - 11x \Rightarrow 10x^2 + 23x - 5 = 0$

substitute

**ANSWER**  $B(-2.5, -5.5)$

IDEA 05

## Practise: Line Meets Curve

My questions first, then selected practice, then real past papers.

**Q42** DR ESLAM Line meets curve

CH 3 • TOPIC 2 4 marks

Find where  $y = x + 1$  meets  $x^2 + y^2 = 25$ .

## SELECTED PRACTICE

**Q43****PRACTICE**

Practice • Very Hard

CH 3 • TOPIC 2 6 marks

- 4 The curve with equation  $x^2 - x + y^2 = 10$  and the straight line with equation  $x - y = -4$  intersect at the points  $A$  and  $B$ .

Work out the exact length of  $AB$ .

Show your working clearly and give your answer in the form  $\frac{\sqrt{a}}{2}$  where  $a$  is an integer.

**(6 marks)**

**Q44****PRACTICE**

Practice · Very Hard

**CH 3 · TOPIC 2 6 marks**

- 5 The curve with equation  $y = (10x - 3)(x + 1)$  and the line with equation  $y - 6x = 0$  intersect at the points  $A$  and  $B$ .

Find the coordinates of the midpoint of  $AB$ .

Show your working clearly.

**(6 marks)**

## PAST PAPERS

**Q45****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q22

**CH 3 • TOPIC 2 5 marks**

- 22 The line with equation  $y = x + 2$  intersects the curve with equation  $x^2 + y^2 - 2y = 24$  at the points  $A$  and  $B$ .

Find the coordinates of  $A$  and  $B$ .  
Show clear algebraic working.

(....., .....)

(....., .....)

(Total for Question 22 is 5 marks)

**Q46****PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q19

**CH 3 • TOPIC 2 6 marks**

- 19 The straight line **L** has equation  $x - y = 3$   
The curve **C** has equation  $3x^2 - y^2 + xy = 9$

**L** and **C** intersect at the points *P* and *Q*.

Find the coordinates of the midpoint of *PQ*.  
Show clear algebraic working.

(..... , .....)

**(Total for Question 19 is 6 marks)**

**Q47****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q18

**CH 3 • TOPIC 2 5 marks**

- 18** The line with equation  $2y = x + 1$  intersects the curve with equation  $3y^2 + 7y + 16 = x^2 - x$  at the points  $A$  and  $B$

Find the coordinates of  $A$  and the coordinates of  $B$   
Show clear algebraic working.

(..... , ..... ) and (..... , ..... )

**(Total for Question 18 is 5 marks)**

**Q48****PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q21

**CH 3 • TOPIC 2 5 marks**

21 Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(..... , .....)

(..... , .....)

**(Total for Question 21 is 5 marks)**

**Q49****PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q21

**CH 3 • TOPIC 2 5 marks**

- 21 The line with equation  $x + 2y = 5$  intersects the curve with equation  $x^2 + 3y^2 = 13$  at the points  $A$  and  $B$

Find the coordinates of  $A$  and the coordinates of  $B$   
Show clear algebraic working.

(..... , .....)

(..... , .....)

(Total for Question 21 is 5 marks)

**Q50****PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q22

**CH 3 • TOPIC 2 5 marks**

**22** The straight line **L** has equation  $x + y = 5$

The curve **C** has equation  $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.

Show clear algebraic working.

(..... , ..... ) (..... , ..... )

**(Total for Question 22 is 5 marks)**

**Q51****PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q28

**CH 3 • TOPIC 2 6 marks**

- 28 The curve **C** has equation  $x^2 + y^2 = d - 11x$  where  $d$  is an integer.  
The line **L** has equation  $y = 3x + e$  where  $e$  is an integer.

**C** and **L** intersect at the point  $A$  and at the point  $B$

The coordinates of  $A$  are  $(0.2, 2.6)$

Work out the coordinates of  $B$   
Show your working clearly.

( ..... , ..... )

(Total for Question 28 is 6 marks)

# ALL The Whole Topic on One Page

Tear this page out before the exam.

**ZONE MID**  
average

**ZONE m**  
rise  $\div$  run

**ZONE LEN**  
Pythagoras

**ZONE  $\perp$**   
 $m_1 m_2 = -1$

- 01 Midpoint / length**      Average; Pythagoras; missing end: same step again.
- 02 Equation**               $m$  first, then  $y - y_1 = m(x - x_1)$ ; rearrange to the form.
- 03  $\parallel$  and  $\perp$**               Same  $m$ ;  $-\frac{1}{m}$ ; bisector through the midpoint.
- 04 Shapes**                  Use right angles, parallel sides, equal diagonals.
- 05 Line meets curve**        Substitute, solve, pair the points.

**BOTTOM LINE** Sketch it  $\rightarrow$  gradient first  $\rightarrow y - y_1 = m(x - x_1) \rightarrow$  perpendicular: flip and change sign.

# Answers

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Coordinate Geometry • for the teacher. Every answer checked with a computer algebra system.

## IDEA 01 • MIDPOINT AND LENGTH

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- |           |                                              |                                               |
|-----------|----------------------------------------------|-----------------------------------------------|
| <b>Q1</b> | Dr Eslam • Midpoint                          | $(2, 2); \sqrt{136} = 2\sqrt{34}$             |
| <b>Q2</b> | Practice • Easy                              | $(4, 5.5)$                                    |
| <b>Q3</b> | Practice • Easy                              | $Q(4, 16)$                                    |
| <b>Q4</b> | Practice • Medium                            | $C - A = 4(B - A): p = 8, q = 10$             |
| <b>Q5</b> | Practice • Hard                              | $P(23, 46)$                                   |
| <b>Q6</b> | Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q6 | $a = 22, b = -49$                             |
| <b>Q7</b> | Edexcel 4MA1 • June 2022 • Paper 1H • Q10    | $(a) 17 \text{ cm } (b) 13 - 6 < 17 < 13 + 6$ |
| <b>Q8</b> | Edexcel 4MA1 • November 2025 • Paper 2H • Q1 | $M(6, 10)$                                    |

## IDEA 02 • GRADIENT AND THE EQUATION OF A LINE

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- Q9** Dr Eslam • Line through two points  $m = 3: y = 3x$
- Q10** Practice • Easy  $-\frac{5}{2}$
- Q11** Practice • Easy 2
- Q12** Practice • Hard (a) (3, 6.5) (b)  $y = \frac{1}{2}x + 5$ : at 100,  $y = 55 \neq 56$   
— no
- Q13** Practice • Hard 2
- Q14** Edexcel 4MA1 • January 2020 • Paper 1H • Q1 (a) (9, -1.5) (b) -3 (c)  $2 - 300 = -298 \neq -302$ :  
no
- Q15** Edexcel 4MA1 • January 2021 • Paper 1H • Q13 (a) -3 (b)  $\frac{1}{4}$
- Q16** Edexcel 4MA1 • January 2022 • Paper 1H • Q10 (a)  $-\frac{7}{2}$  (b) (0, 5)
- Q17** Edexcel 4MA1 • May 2023 • Paper 1H • Q13  $y = -\frac{5}{2}x - \frac{1}{2}$

### IDEA 03 • PARALLEL AND PERPENDICULAR

- Q18** Dr Eslam • Perpendicular  $m = -2: y = -2x + 5$
- Q19** Practice • Medium  $-\frac{3}{2} \times \frac{3}{4} \neq -1$
- Q20** Practice • Hard  $MN$  and  $PR$  both have gradient  $\frac{1}{2}$
- Q21** Edexcel 4MA1 • January 2020 • Paper 2H • Q19 (a) e.g.  $y = -4x + 1$  (b)  $5x - 3y + 42 = 0$
- Q22** Edexcel 4MA1 • May 2021 • Paper 2H • Q17  $(20, 0)$
- Q23** Edexcel 4MA1 • June 2022 • Paper 2HR • Q20  $y = -\frac{1}{2}x + 4$
- Q24** Edexcel 4MA1 • June 2023 • Paper 2HR • Q25  $j = -10, k = -1: (-2, 3)$
- Q25** Edexcel 4MA1 • November 2023 • Paper 1H • Q22 26.3 cm
- Q26** Edexcel 4MA1 • June 2025 • Paper 2HR • Q24  $y = -3x + 7$
- Q27** Edexcel 4MA1 • November 2025 • Paper 1H • Q20  $y = -\frac{1}{4}x + 3$

## IDEA 04 • SHAPES ON A GRID

- Q28** Dr Eslam • Right angle  $m_{AB} = \frac{1}{2}, m_{BC} = -2: \frac{6}{k-4} = -2, k = 1$
- Q29** Practice • Medium e.g.  $A(0, 4), B(0, -4), C(7, 0)$
- Q30** Practice • Hard  $C(30, 9), D(13, -8)$
- Q31** Edexcel 4MA1 • January 2020 • Paper 1H • Q22  $j = -2, k = 11$
- Q32** Edexcel 4MA1 • January 2020 • Paper 2HR • Q24  $A(5, 0), B(-3, -16): \text{area } 40$
- Q33** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q22  $(0, -\frac{89}{7})$
- Q34** Edexcel 4MA1 • January 2021 • Paper 1HR • Q22  $5y + 3x = 39$
- Q35** Edexcel 4MA1 • January 2021 • Paper 2H • Q18 (a)  $a = 2.5, b = 14, c = -9.5$  (b) 65 cm
- Q36** Edexcel 4MA1 • November 2021 • Paper 1H • Q20  $C(-1, 8): \text{area } 30.5$
- Q37** Edexcel 4MA1 • January 2022 • Paper 2HR • Q22  $p = -6, q = -2$
- Q38** Edexcel 4MA1 • June 2022 • Paper 2H • Q22  $B(3, 6.5), D(-3, -2.5)$
- Q39** Edexcel 4MA1 • June 2023 • Paper 2H • Q24  $D(4.6, 8)$
- Q40** Edexcel 4MA1 • June 2024 • Paper 2HR • Q21  $8x + 3y - 39 = 0$
- Q41** Edexcel 4MA1 • June 2025 • Paper 1H • Q25  $3y = x - 3$

### IDEA 05 • LINE MEETS CURVE

- Q42** Dr Eslam • Line meets curve  $2x^2 + 2x - 24 = 0$ :  $(3, 4)$  and  $(-4, -3)$
- Q43** Practice • Very Hard  $A(-2, 2), B(-1.5, 2.5)$ :  $\frac{\sqrt{2}}{2}$
- Q44** Practice • Very Hard  $(-0.05, -0.3)$
- Q45** Edexcel 4MA1 • January 2020 • Paper 2H • Q22  $(3, 5)$  and  $(-4, -2)$
- Q46** Edexcel 4MA1 • November 2021 • Paper 2H • Q19  $(-0.5, -3.5)$
- Q47** Edexcel 4MA1 • January 2022 • Paper 2HR • Q18  $(-3, -1)$  and  $(27, 14)$
- Q48** Edexcel 4MA1 • June 2023 • Paper 2HR • Q21  $(-1.5, -2)$  and  $(\frac{2}{3}, \frac{7}{3})$
- Q49** Edexcel 4MA1 • November 2023 • Paper 2H • Q21  $(1, 2)$  and  $(\frac{23}{7}, \frac{6}{7})$
- Q50** Edexcel 4MA1 • June 2024 • Paper 2H • Q22  $(-3, 8)$  and  $(9, -4)$
- Q51** Edexcel 4MA1 • November 2025 • Paper 1H • Q28  $d = 9, e = 2$ :  $B(-2.5, -5.5)$